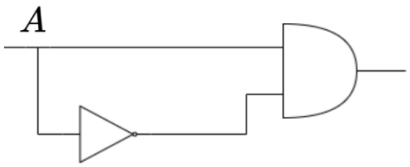
Tutorial 04 - Boolean Algebra

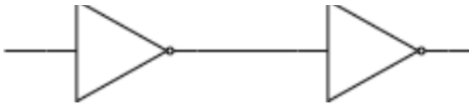
1. Write boolean expressions to represent the following logic gates. (Ex: Y = A + B')
   a. AND
   b. OR
   c. NOT
   d. NAND
   e. NOR
   f. XOR
   g. XNOR

2. Derive boolean expressions for the following circuit diagrams.

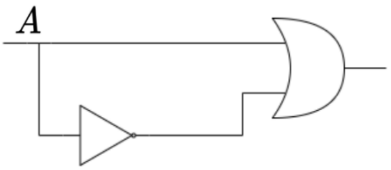
a.



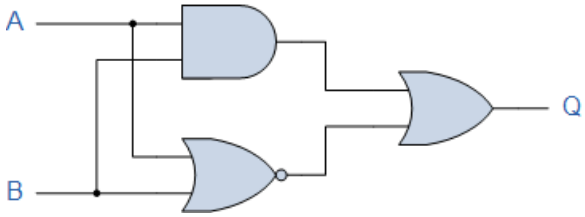
b.



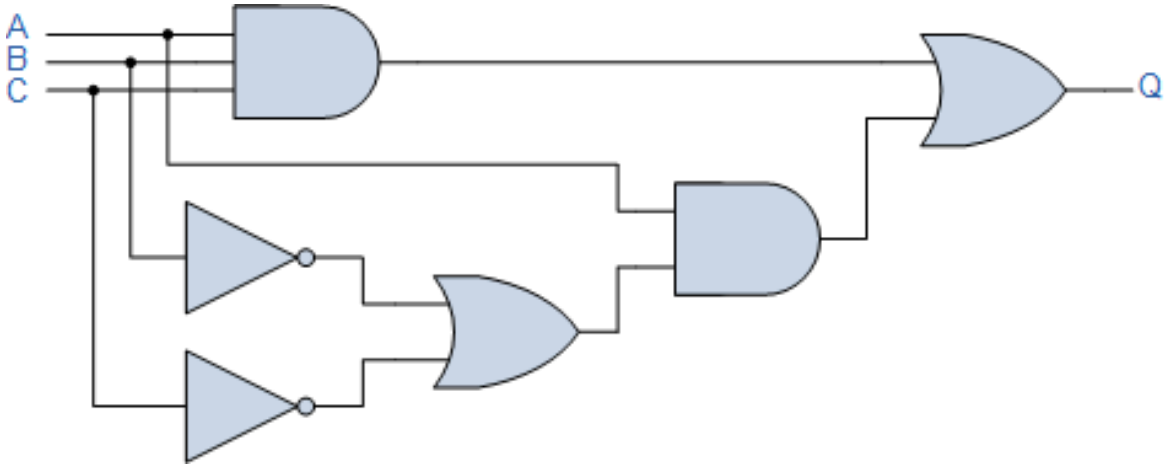
c.



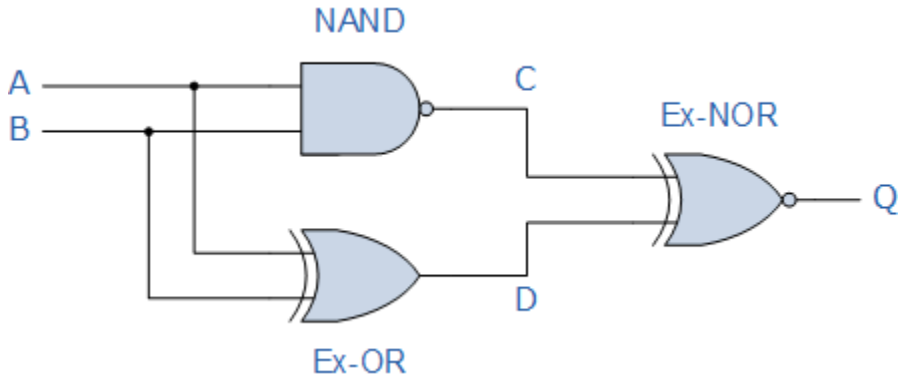
d.



e.



f.



3. State the boolean algebra laws respect to AND and OR gates.

4. State whether the following statements are **True** or **False**
   a. A NAND gate is equivalent to an inversion followed by a NOR
   b. A NAND gate is equivalent to an inversion followed by an OR
   c. A NOR gate is equivalent to an inversion followed by an AND
   d. A NOR gate is equivalent to an inversion followed by an OR

5. Draw logic circuits for the following boolean expressions and simplify using laws as much as possible.

   a. $C + \overline{BC}$:

   b. $\overline{AB}(\overline{A} + B)(\overline{B} + B)$:

   c. $A\,B\,C + \overline{A}\,B + A\,B\,\overline{C}$

6. How many gates would be required to implement the following Boolean expression before simplification? **XY + X(X + Z) + Y(X + Z)**

7. How many gates would be required to implement the following Boolean expression after simplification?